ENTERPRISE RISK MANAGEMENT DASHBOARD — PROOF OF CONCEPT

Comprehensive Technical & Business Documentation

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1. What Has Been Done for the POC

This Proof of Concept (POC) demonstrates a fully integrated, formula-driven Enterprise Risk Management (ERM) Dashboard designed for a large Indian public-sector bank. The POC has been built in three tightly coupled layers — data, analytics, and visualization — with every KRI value traceable back to a single source of truth.

1.1 Data Architecture & Excel Model

The backbone of the POC is a 20-sheet Microsoft Excel workbook (Enterprise_Risk_Dashboard_Formula_Driven.xlsx) built entirely on formula-driven logic. No KRI value is hardcoded into a dashboard cell; all metrics auto-recalculate when source data changes.

- Monthly_Data sheet: Central repository of 65+ raw metrics spanning 12 months (May-2025 to Apr-2026) across all risk categories.
- 20 individual sheets covering Pillar 1, Pillar 2, and Pillar 3 risk domains.
- 200 transaction/position records per risk category — 200 credit exposures, 200 trading positions, 200 operational incidents, 200 liquidity records, and so on.
- KRI_Thresholds sheet: A master reference of 40 KRIs with Green/Amber/Red thresholds, regulatory references, data sources, and owners.
- Monthly_KRI_Trend sheet: 12-month trend series for 40 KRIs — all values computed via Excel formulas from Monthly_Data.
- KRI_Master_Summary sheet: Single-page risk cockpit showing current values, RAG status, MoM change, and breach flag for all 69 KRIs.

1.2 Backend API (FastAPI / Python)

A FastAPI-based Python backend (main.py) serves the Excel data as RESTful JSON APIs. Key features implemented in this POC include:

- Auto-loading and in-memory caching of all 20 Excel sheets on startup.
- 30+ API endpoints covering KRI summary, trend series, pillar-wise risk details, and summary aggregations.
- Intelligent unit classification (percentage, integer, currency, score) and display formatting.
- Live KRI Ticker: a scrolling midnight-dark styled bar injected into every HTML page showing real-time KRI values with RAG color coding.
- CORS middleware for cross-origin requests and an HTML theme injection middleware that patches Chart.js defaults for dark-mode rendering.
- Optional filters on most endpoints (by segment, RAG status, scenario, dimension, etc.).

1.3 Frontend Web Dashboard

A static HTML/JavaScript frontend served via FastAPI's StaticFiles mechanism provides an interactive browser dashboard with:

- Pillar-wise navigation across Credit, Market, Operational, Capital, Liquidity, IRRBB, Concentration, Climate, Cyber, Fraud, Third Party, Reputational, Strategic, and Pillar 3 Disclosure risk views.
- RAG traffic-light scorecards, KRI trend line charts, bar charts, and donut charts.
- Animated, real-time KRI ticker bar scrolling all 40+ KRIs with current values, trend arrows, and RAG badges.
- Responsive layout with dark-themed cards and high-contrast data visualization.

1.4 Scope Coverage

Pillar	Risk Categories Covered	Sheets	KRIs
Pillar 1	Credit Risk, Market Risk, Operational Risk, Capital Adequacy	5	18
Pillar 2	Liquidity, IRRBB, Concentration, Climate, Cyber, Fraud, Third Party, Reputational, Strategic	12	46
Pillar 3	Capital Disclosure, Market Discipline Disclosure	2	3 (+ qualitative)
Supporting	Monthly Data, KRI Thresholds, KRI Trend, TOC	4	Master
Total	All Basel III / RBI Pillars	20	69

2. What the Solution Does

The Enterprise Risk Management Dashboard is an end-to-end digital risk intelligence platform that consolidates all regulatory and internal risk metrics of a large bank into a single, interactive, formula-driven interface. It automates the manual process of compiling risk reports from multiple source systems into a cohesive management information system.

2.1 Core Capabilities

- Real-Time Risk Monitoring: Tracks 69 Key Risk Indicators across 14 risk categories on a daily, weekly, fortnightly, monthly, quarterly, or annual basis as appropriate to each metric.
- Automated RAG Status Engine: Every KRI is automatically classified as Green (within tolerance), Amber (early warning zone), or Red (breach/critical) by comparing the current value against pre-defined threshold bands.
- Month-on-Month Trend Analysis: Calculates MoM change for every KRI and provides 12-month rolling averages and 3-month rolling averages to identify directional trends.
- Regulatory Compliance Dashboard: Maps every KRI to its regulatory reference — RBI IRAC, Basel III, FEMA, CERT-In, Ind AS 109, BCBS 239, and more — providing one-click compliance status.
- Pillar 3 Disclosure Tracker: Monitors completeness and timeliness of all Pillar 3 disclosures including capital adequacy, market discipline, and ICAAP submissions.
- Drill-Down Capability: Each risk category has a transaction-level register (200 records) enabling root-cause investigation from KRI to individual exposure, position, incident, or vendor.
- Formula-Driven Architecture: Changing any number in the Monthly_Data sheet propagates automatically through all 40 KRIs, all 12 months of trend data, all RAG statuses, and all dashboard visualizations — zero manual recalculation required.
- API-First Design: Every data element is available as a REST API, enabling integration with existing bank systems (CBS, Treasury, ORMS, FMS, DLP tools).

2.2 User Personas Served

Persona	What They Get from the Dashboard
Chief Risk Officer (CRO)	Single-page KRI Master Summary with 69 metrics, RAG status, and breach flags
Credit Risk Head	Loan book details — NPA buckets, ECL, PD/LGD, SMA flags, PCR, top-20 concentrations
Market Risk Head	Trading book VaR, SVaR, IRC, DV01, FX open positions, Greeks, MTM P&L

Persona	What They Get from the Dashboard
Operational Risk Head	Loss event register, RCSA scores, KCI breaches, BEL estimates, action items
ALM / Treasury Head	LCR, NSFR, HQLA, gap analysis, NII sensitivity, EVE, duration gap, rate shock scenarios
CFO	CRAR, CET-1, Tier-1, RWA breakdown, leverage ratio, capital buffers, ICAAP headroom
CISO	Cyber incidents, MTTD, MTTR, CVSS scores, unpatched vulnerabilities, CERT-In filings
Risk Analytics Team	Full KRI trend series, threshold reference table, formula documentation, data validation
Board / Audit Committee	High-level Pillar 3 disclosure completeness, key breach summary, MoM improvement trends

3. Explanation of Each Sheet

3.1 Monthly_Data — Master Source Sheet

This is the single source of truth for the entire dashboard. All KRI calculations in the Monthly_KRI_Trend sheet and subsequently the KRI_Master_Summary are derived from this sheet via Excel formulas. No values in the downstream sheets are hardcoded.

Column	Meaning
Metric	Full name of the raw data metric (e.g., Gross NPA Amount ₹ Cr, Gross Advances ₹ Cr)
Unit	Unit of measurement (₹ Cr, %, Count, etc.)
May-25 … Apr-26	12 monthly raw data values for each metric — these are the blue-text hardcoded inputs
Source	The originating risk sheet or system (e.g., P1_Credit_Risk, P2_Liquidity_Risk)
Formula / Note	Describes how the raw metric is sourced from core banking systems (e.g., SUMIF, SUM formulas)

The sheet is organized into sections: Credit Risk, Market Risk, Operational Risk, Capital Adequacy, Liquidity Risk, IRRBB, Concentration Risk, Climate Risk, Cyber/Fraud, Third Party/Reputational, and Strategic Risk. Each section contains 3-8 raw metrics needed to calculate the KRIs for that category.

3.2 KRI_Master_Summary — Enterprise Risk Cockpit

This is the primary management dashboard — a single-page summary of all 69 KRIs. Every value is formula-driven from Monthly_KRI_Trend.

Column	Meaning
#	Sequential KRI number (1–69)
Risk Category	Pillar and sub-category (e.g., Pillar 1 – Credit Risk)
KRI Name	Full name of the Key Risk Indicator
Current Value	Latest month's value, pulled via formula from Monthly_KRI_Trend

Column	Meaning
Threshold (Green)	The range within which the metric is considered healthy
Threshold (Amber)	The early-warning range indicating risk requires attention
Threshold (Red)	The breach threshold requiring escalation and management action
Unit	Unit of the KRI (%, ₹ Cr, Count, Score, Years, etc.)
Frequency	Monitoring frequency: Daily, Weekly, Monthly, Quarterly, Annual
RAG Status	Auto-calculated color: Green = safe, Amber = watch, Red = breach
MoM Change	Month-on-month change percentage compared to prior month
Data Source	Primary system of record (CRILC, Treasury System, ORMS, ALM, SIEM, FMS, etc.)
Regulatory Ref	Applicable regulation or standard (RBI IRAC, Basel III, Ind AS 109, etc.)
KRI_Score	Numeric score: 1 = Green, 2 = Amber, 3 = Red — used for aggregation
Breach_Flag	Text flag: OK, WATCH, or YES — quickly identifies KRIs needing action

3.3 KRI_Thresholds — Reference Table

Contains the definitional framework for all 40 primary KRIs. It serves as both a governance document and a technical reference for the threshold logic engine.

Column	Meaning
KRI_ID	Unique identifier (KRI001 to KRI040)
Pillar	P1, P2, or P3
Risk_Type	Sub-category: Credit, Market, OpRisk, Capital, Liquidity, IRRBB, Concentration, Climate, Cyber, Fraud, ThirdParty, Reputational, Disclosure
KRI_Name	Official name as used across all sheets
Unit	Measurement unit
Green_Min / Green_Max	Lower and upper bounds of the Green (safe) zone

Column	Meaning
Amber_Min / Amber_Max	Lower and upper bounds of the Amber (watch) zone
Red_Trigger	The specific threshold value or direction that triggers a Red status
Direction	Lower Better (lower values are safer) or Higher Better (higher values are safer)
Frequency	Measurement cadence
Regulatory_Ref	Authoritative regulation: RBI IRAC, Basel III, FEMA, Ind AS 109, CERT-In, BCBS 239, NGFS, etc.
Data_Source	Source system: CRILC/Finacle, Treasury System, GRC Tool, ALM System, SIEM, FMS, DLP Tool, etc.
Formula_Logic	Plain-English formula (e.g., NPA Amt/Gross Adv *100, HQLA/30D Net Cash Outflow *100)
Owner	Responsible function (Credit Risk Head, Market Risk Head, CFO, CISO, etc.)
Review_Cycle	How often thresholds are reviewed (Monthly, Quarterly, Annual)

3.4 Monthly_KRI_Trend — 12-Month Time Series

This sheet contains month-by-month values for 40 KRIs from May-2025 to Apr-2026. All values are Excel formulas referencing Monthly_Data — no hardcoded numbers appear here (except the threshold columns).

Column	Meaning
#	KRI sequence number
KRI Name	Name of the KRI
Unit	Unit of measurement
Direction	Lower Better or Higher Better — determines trend interpretation
Red Thresh / Amber Thresh	Threshold values for conditional formatting and chart reference lines
May-25 to Apr-26	Monthly KRI values — all are Excel formulas pointing to Monthly_Data
12M Avg	Simple average of all 12 monthly values

Column	Meaning
Apr-26 (Current)	Latest month's value — used by KRI_Master_Summary
3M Avg	Average of the last 3 months (Feb-Mar-Apr)
MoM Change	(Apr-26 – Mar-26) / Mar-26 — percentage change month on month
Trend	Text indicator: ↑ Improving, ↓ Worsening, or → Stable

3.5 P1_Credit_Risk — Loan Book Register (200 Records)

Contains individual loan-level data for 200 borrowers across Retail, Corporate, MSME, Priority Sector, and Infrastructure segments. This is the base data from which all credit KRIs are derived.

Column	Meaning
Loan_ID	Unique loan identifier (LN20000 to LN20199)
Borrower	Anonymized borrower name
Segment	Retail, Corporate, MSME, Priority Sector, Infrastructure
Loan_Type	Home Loan, Working Capital, MSME Loan, Term Loan, etc.
Sanction_Amt_Cr	Original sanctioned amount in ₹ Crore
Outstanding_Cr	Current outstanding balance in ₹ Crore
DPD_Days	Days Past Due — number of days the account is overdue
NPA_Category	Standard, SMA-0, SMA-1, SMA-2, Sub-Standard, Doubtful, Loss
PD_%	Probability of Default — the estimated likelihood the borrower will default
LGD_%	Loss Given Default — percentage of exposure expected to be lost if default occurs
EAD_Cr	Exposure at Default — outstanding amount at the time of default
ECL_Cr	Expected Credit Loss = PD% × LGD% × EAD — IndAS 109 provision requirement
CRAR_%	Capital to Risk-Weighted Assets Ratio — solvency indicator at borrower level
Collateral	Type of security: Property, Stocks, Plant & Machinery, Guarantee, etc.

Column	Meaning
Risk_Rating	Internal credit rating: AAA to D
SMA_Flag	Yes/No — indicates Special Mention Account status
PCR_%	Provision Coverage Ratio — provisions held as a % of gross NPA
Review_Date	Next account review date
Sector	Industry sector: Retail, Shipping, Agriculture, Infrastructure, etc.
Geography	State of borrower location
Status	Normal, Watch, or Stressed
RAG	Green, Amber, or Red — summary risk status of the loan

3.6 P1_Market_Risk — Trading Book Positions (200 Records)

Records individual trading and investment book positions across five desks: Rates, Equity, Credit, FX, and Commodity. Used to compute VaR, SVaR, IRC, and Greeks-based KRIs.

Column	Meaning
Position_ID	Unique position identifier
Trade_Date	Date the position was entered
Desk	Trading desk: Rates, Equity, Credit, FX, Commodity
Asset_Class	Government Bond, Corporate Bond, Equity, FX Forward, Commodity, Derivative
Instrument	Specific instrument type
Currency	Currency of denomination: INR, USD, EUR, JPY, SGD, GBP, CHF
Face_Value_Cr	Notional/face value of the position in ₹ Crore
MTM_Cr	Mark-to-Market value in ₹ Crore
VaR_1D_Cr	1-day 99% Value at Risk in ₹ Crore
SVaR_Cr	Stressed VaR (historical stress period) in ₹ Crore
IRC_Cr	Incremental Risk Charge for credit migration and default risk

Column	Meaning
ES_99_Cr	Expected Shortfall at 99% confidence level
Duration	Modified duration in years
DV01_Lakhs	Dollar Value of 01 — P&L impact of 1 basis point move in ₹ Lakh
Greeks_Delta	Delta — sensitivity of position value to underlying price movement
Limit_Cr	Applicable trading limit in ₹ Crore
Limit_Used_%	Current position as a percentage of the limit
Breach_Flag	Yes/No — indicates if any limit is breached
Rating	Credit rating of the instrument
Maturity_Date	Instrument maturity/expiry date
Status	Normal, Watch, or Stressed
RAG	Green, Amber, or Red — overall position risk status

3.7 P1_Operational_Risk — Loss Event Register (200 Incidents)

Maintains the operational risk loss event database across all business units, covering the seven Basel III event types: Internal Fraud, External Fraud, Employment Practices, Clients/Products, Damage to Assets, Business Disruption, and Execution/Delivery.

Column	Meaning
Incident_ID	Unique incident reference
Date	Date the incident occurred
Business_Unit	HR, Trade Finance, Retail Banking, Treasury, Operations, IT, Compliance, etc.
Risk_Category	People Risk, Internal Fraud, Technology Risk, Process Risk, External Fraud, etc.
Event_Type	Fraud, System Failure, Natural Disaster, Regulatory Breach, Data Breach, etc.
Gross_Loss_Lakhs	Total financial loss before recoveries in ₹ Lakh
Net_Loss_Lakhs	Loss after insurance and other recoveries

Column	Meaning
Recovery_Lakhs	Amount recovered from insurance or third parties
Frequency	Number of similar events in the period
RCSA_Score	Risk and Control Self-Assessment score for the associated process
KRI_Breach	Yes/No — was a KRI threshold breached in connection with this incident?
RCA_Done	Yes/No/In Progress — status of Root Cause Analysis
Control_Adequacy	Weak, Adequate, or Strong — effectiveness of controls at time of event
Residual_Risk	Low, Medium, or High — residual risk level after controls
BEL_Estimate_Lakhs	Basic Expected Loss estimate for capital provisioning purposes
Action_Owner	Designated owner responsible for remediation
Target_Date	Target date for closure of remediation actions
Op_Capital_Cr	Operational risk capital charge attributed to this incident in ₹ Crore
Status	Open or Closed
RAG	Green, Amber, or Red

3.8 P1_Capital_Adequacy — CRAR & RWA Trend (Quarterly)

Contains 12 quarters of capital adequacy data from Q1-FY24 to Q4-FY26. All values represent the bank's regulatory capital position under Basel III / RBI norms.

Column	Meaning
Quarter	Fiscal quarter (Q1-FY24 to Q4-FY26)
CET1_Cr	Common Equity Tier 1 capital in ₹ Crore — highest quality capital
Tier1_Cr	Total Tier 1 capital = CET1 + Additional Tier 1 (AT1)
Tier2_Cr	Tier 2 capital (subordinated debt, loan loss provisions)
Total_Capital_Cr	Total regulatory capital = Tier 1 + Tier 2
Credit_RWA_Cr	Risk-Weighted Assets for credit risk exposures

Column	Meaning
Market_RWA_Cr	RWA for market risk (trading book)
Op_RWA_Cr	RWA for operational risk
Total_RWA_Cr	Sum of all three RWA components
CET1_%	CET1 Ratio = CET1 / Total RWA — minimum 8% as per RBI
Tier1_%	Tier 1 Ratio — minimum 9.5% (10% for D-SIBs) as per RBI
CRAR_%	Capital to Risk Weighted Assets Ratio — minimum 11.5% as per RBI (+ D-SIB buffer)
CCB_%	Capital Conservation Buffer — must be ≥2.5% per Basel III
D-SIB_Buffer_%	Additional D-SIB surcharge for systemically important banks (0.6% for SBI)
Leverage_Ratio_%	Tier 1 Capital / Total Exposure — minimum 4% as per RBI
Capital_Buffer_Cr	Excess capital above minimum requirement
ICAAP_Headroom_Cr	Available headroom under Internal Capital Adequacy Assessment Process
RAG	Green/Amber/Red status based on CRAR bands

3.9 P2_Liquidity_Risk — LCR/NSFR/Gap Analysis (200 Records)

Contains cash flow gap analysis records across various time buckets from overnight to 5+ years, along with LCR, NSFR, SLR, and CRR ratios under different stress scenarios.

Column	Meaning
Record_ID	Unique record identifier
Date	Date of the liquidity observation
Time_Bucket	Maturity bucket: Overnight, 2-7 Days, 8-14 Days, 1-2M, 2-3M, 3-6M, 6M-1Y, 1Y-3Y, 3Y-5Y, >5Y
Inflow_Cr	Cash inflows expected in the bucket (₹ Crore)
Outflow_Cr	Cash outflows expected in the bucket (₹ Crore)

Column	Meaning
Net_Gap_Cr	Net cash flow = Inflow – Outflow (positive = surplus, negative = deficit)
Cumul_Gap_Cr	Cumulative gap from the first bucket to the current bucket
LCR_%	Liquidity Coverage Ratio = HQLA / 30-day Net Cash Outflow × 100 — minimum 100%
NSFR_%	Net Stable Funding Ratio = ASF / RSF × 100 — minimum 100%
SLR_%	Statutory Liquidity Ratio — minimum 18% of NDTL as per RBI
CRR_%	Cash Reserve Ratio — minimum 4% of NDTL as per RBI
HQLA_Cr	High Quality Liquid Assets — unencumbered, liquid assets available for stress outflows
HQLA_L1_Cr	Level 1 HQLA (cash, central bank reserves, G-Secs) — no haircut
HQLA_L2_Cr	Level 2 HQLA (selected corporate bonds, equity) — subject to 15-50% haircut
Net_Cash_Outflow_30D_Cr	Projected net cash outflows over 30 days under stress
Stress_Scenario	Scenario applied: Base, Moderate Stress, or Severe Stress
NSFR_RSF_Cr	Required Stable Funding — amount needing to be funded stably
NSFR_ASF_Cr	Available Stable Funding — stable funding actually available
Status	Normal, Watch, or Stressed
RAG	Green, Amber, or Red

3.10 P2_IRRBB — Interest Rate Risk in Banking Book (200 Records)

Captures the bank's interest rate risk sensitivity under six rate shock scenarios: +100bps, -100bps, +200bps, -200bps, +300bps, Bull Flat, Bear Flat, and Parallel Shift scenarios per RBI IRRBB guidelines.

Column	Meaning
Record_ID	Unique record identifier
Date	Date of the IRRBB measurement
Rate_Shock_bps	Interest rate shock applied in basis points (e.g., +200, -100)

Column	Meaning
Asset_Repricing_Cr	Value of assets repricing within the relevant time horizon
Liability_Repricing_Cr	Value of liabilities repricing within the relevant time horizon
Repricing_Gap_Cr	Asset repricing minus liability repricing — positive means assets reprice faster
NII_Impact_Cr	Change in Net Interest Income due to the rate shock (₹ Crore)
NII_Impact_%	NII change as % of base NII — key KRI threshold: must be > -5%
EVE_Impact_Cr	Change in Economic Value of Equity — present value impact of rate shock
EVE_Impact_Pct_Tier1	EVE change as % of Tier 1 capital — must stay above -8% per Basel IRRBB
Duration_Assets_Yrs	Weighted average duration of the asset portfolio in years
Duration_Liab_Yrs	Weighted average duration of the liability portfolio in years
Duration_Gap_Yrs	Duration of Assets minus Duration of Liabilities — KRI threshold <1.5 yrs
MCLR_1Y_%	Marginal Cost of Funds-based Lending Rate for 1-year tenor
Base_Rate_%	Base lending rate
Deposit_Rate_%	Weighted average deposit rate
Yield_Portfolio_%	Yield on the banking book investment portfolio
Delta_NII_Parallel_Up/Down/Steepener	NII sensitivity under three specific rate scenarios
Scenario	Named scenario: +200bps, +300bps, Bull Flat, Bear Flat, Steepener, Flattener
Status / RAG	Risk status indicators

3.11 P2_Concentration_Risk — Sector/Geo/Borrower Analysis (200 Records)

Analyses concentration risk across four dimensions: Sector, Geography, Product, and Maturity. Each record captures an exposure dimension and its regulatory utilisation.

Column	Meaning
Dimension	Sector, Geography, Product, Maturity, or Borrower Group
Category	Specific category (e.g., Steel sector, Maharashtra geography, Term Loan product)
Exposure_Cr	Total outstanding exposure to this category in ₹ Crore
Total_Advances_Cr	Total advances book for computing concentration percentage
Conc_%	Concentration ratio = Exposure / Total Advances × 100
Regulatory_Limit_%	RBI-prescribed limit for this dimension (e.g., 40% for sector concentration)
Utilisation_%	Actual concentration as % of the regulatory limit
HHI_Index	Herfindahl-Hirschman Index — measures portfolio concentration (0=fully diversified, 1=monopoly)
Top5_Exposure_Cr	Sum of exposures to top 5 categories in the dimension
Top10_Exposure_Cr	Sum of exposures to top 10 categories
Single_Borrower_Cap_%	RBI ceiling: single borrower may not exceed 15% of capital funds
Group_Borrower_Cap_%	RBI ceiling: borrower group may not exceed 25% of capital funds
Breach_Flag	Yes/No — whether regulatory concentration limit is breached
Action_Required	No Action, Scheduled Review, or Immediate Review
Status	Low, Medium, High
RAG	Green, Amber, or Red

3.12 P2_Climate_Risk — Physical & Transition Risk (200 Records)

Assesses climate-related financial risk for 200 borrowers/assets aligned to TCFD (Task Force on Climate-related Financial Disclosures) and RBI's climate risk framework.

Column	Meaning
Asset_ID	Unique asset/borrower identifier

Column	Meaning
Physical_Risk_Score	Score 1–10 measuring exposure to physical climate hazards (floods, droughts, cyclones, heat stress)
Transition_Risk_Score	Score 1–10 measuring exposure to policy, technology, market, and reputational transition risks
Overall_Climate_Score	Composite climate risk score = weighted average of physical and transition scores
Flood_Prob_% / Drought_Prob_%	Probability of flood/drought events affecting the borrower's location
Cyclone_Prob_% / Heat_Stress_Score	Cyclone probability and heat stress intensity score
Carbon_Intensity_tCO2/Cr	GHG emissions per ₹ Crore of exposure — portfolio decarbonisation metric
GHG_Scope1/2/3_tCO2	Direct emissions (Scope 1), indirect energy emissions (Scope 2), value chain emissions (Scope 3)
TCFD_Category	TCFD risk classification: Physical-Acute, Physical-Chronic, Transition-Policy, Transition-Technology, Liability
Net_Zero_Aligned	Yes/No/Partial — whether borrower's business plan aligns with net zero targets
ESG_Rating	Borrower's ESG rating from AAA to D
Green_Finance	Yes/No — whether this exposure qualifies as green/sustainable finance
Climate_VaR_Cr	Climate Value at Risk — potential loss due to climate events in ₹ Crore
Stranded_Asset_Risk	Low/Medium/High — risk of assets becoming stranded due to energy transition
Status / RAG	Risk categorization

3.13 P2_Cyber_Risk — Incident Register (200 Incidents)

Maintains the cybersecurity incident register covering all reportable and internal cyber events, aligned to CERT-In reporting requirements and NIST Cybersecurity Framework.

Column	Meaning
Incident_ID	Unique cyber incident identifier
Severity	Critical, High, Medium, or Low
Attack_Vector	Ransomware, SQL Injection, Phishing, DDoS, Insider Threat, Supply Chain, Zero-Day
Asset_Affected	ATM Network, Core Banking, Mobile App, Internet Banking, Email, Data Centre, Third-Party
System_Type	Internal, External, Hybrid, or Third-Party
Downtime_Hrs	Business downtime caused by the incident in hours
Data_Records_Compromised	Number of customer/internal data records exposed or stolen
Financial_Impact_Lakhs	Direct financial impact in ₹ Lakh
CVE_ID	Common Vulnerabilities and Exposures identifier (e.g., CVE-2022-3694)
CVSS_Score	Common Vulnerability Scoring System score (0–10, with 10 being most critical)
Patch_Status	Patched, In Progress, Overdue, or Not Applicable
CERT_In_Reported	Yes/No — mandatory reporting to CERT-In for qualifying incidents
Response_Time_Hrs	Time from detection to initial response
MTTD_Hrs	Mean Time to Detect — average time from attack initiation to detection
MTTR_Hrs	Mean Time to Respond/Recover — average time from detection to containment
Threat_Actor	Organized Crime, Nation-State, Insider, Hacktivist, Unknown
Vulnerability_Count	Number of known vulnerabilities associated with the incident

3.14 P2_Fraud_Risk — Case Register (200 Cases)

Tracks all fraud incidents detected across banking channels, covering identification, investigation, recovery, and RBI reporting.

Column	Meaning
Case_ID	Unique fraud case identifier
Channel	ATM, Mobile Banking, Internet Banking, NEFT/RTGS, Branch, POS
Fraud_Type	Card Skimming, Cheque Fraud, Identity Theft, Phishing, KYC Fraud, Loan Fraud, Cyber Fraud
Amount_Lakhs	Total fraud amount in ₹ Lakh
Recovered_Lakhs	Amount successfully recovered
Net_Loss_Lakhs	Actual net loss after recovery
Recovery_Rate_%	Recovered / Amount × 100 — key KRI
Detection_Method	How the fraud was identified: SOC Alert, Inter-Bank Alert, Customer Complaint, CBS Monitoring, Manual
Reported_RBI	Yes/No — whether the case has been reported to RBI (mandatory for frauds ≥₹1 Lakh)
FIR_Filed	Yes/No — whether a First Information Report has been filed with law enforcement
Perpetrator	Internal Employee, External Party, Foreign Entity, Organized Crime, Unknown
Resolution_Days	Number of days to resolve/close the case
SuspAcc_Frozen	Yes/No — whether suspicious accounts have been frozen
AML_Flag	Yes/No/Pending — Anti-Money Laundering flag status
Investigation_Status	Under Investigation, Closed, Pending, Referred to Law Enforcement

3.15 P2_Third_Party_Risk — Vendor Risk Register (100 Vendors)

Covers the risk management of 100 critical and non-critical vendors providing outsourced services to the bank, aligned to RBI's Outsourcing Guidelines.

Column	Meaning
Vendor_ID	Unique vendor identifier

Column	Meaning
Service_Category	Regulatory Reporting, Cybersecurity, Cloud, ATM Management, Recovery/BCP, Payment Processing, etc.
Criticality	High, Medium, or Low — based on operational and regulatory criticality
Contract_Value_Cr	Annual contract value in ₹ Crore
SLA_Uptime_%	Contracted Service Level Agreement uptime target
Actual_Uptime_%	Actual uptime delivered
SLA_Breach	Yes/No — whether actual uptime fell below the contracted SLA
Vendor_Risk_Score	Composite risk score 1–5 (lower is better)
Data_Access_Level	PII Access, Financial Data, None, Confidential
TPRA_Score	Third Party Risk Assessment score out of 100
VAPT_Done	Vulnerability Assessment & Penetration Testing status: Done, Scheduled, Overdue
IS_Audit_Done	Information Security audit status
BCP_Tested	Business Continuity Plan test status: Yes, No, Partial, In Progress
RBI_Regulated	Yes/No/N/A — whether the vendor is subject to RBI regulation
Sub_Vendors	Count of sub-contractors (fourth-party risk indicator)
Incident_Count	Number of incidents associated with this vendor in the period
Action_Required	No Action, Scheduled Review, Immediate Review, Terminate

3.16 P2_Reputational_Risk — Sentiment & Complaint Tracker

Tracks reputational risk events including media coverage, social media sentiment, customer complaints, NPS, RBI penalties, and litigation.

Column	Meaning
Category	Technology Failure, Employee Relations, Media Relations, Regulatory Penalty, Legal Action, Customer Service
Sentiment_Score	NLP-based sentiment score 0–100 from media analytics tool (>60 = positive, <50 = negative)

Column	Meaning
Media_Coverage_Count	Number of media articles covering the event
Social_Media_Mentions	Total social media mentions related to the category
Negative_Mentions_%	Percentage of mentions that are negative in sentiment
Customer_Complaints	Number of customer complaints related to the category
Complaints_Resolved_%	Resolution rate of complaints in the period
Resolution_TAT_Days	Average turnaround time to resolve complaints in days
NPS_Score	Net Promoter Score — customer loyalty and advocacy metric
Brand_Trust_Index	Proprietary brand trust index score
RBI_Observations	Number of RBI inspection observations in the period
Penalty_Amt_Cr	Financial penalties levied by RBI or other regulators (₹ Crore)
Parliamentary_Questions	Number of parliamentary questions relating to the bank
Litigation_Count	Number of active litigations
Litigation_Amt_Cr	Total amount at stake in active litigations (₹ Crore)
Reputation_Index	Composite reputational health score

3.17 P2_Strategic_Risk — BSC & KPI Dashboard

Measures strategic objectives and performance using a Balanced Scorecard (BSC) framework across four perspectives: Financial, Customer, Internal Process, and Learning & Growth.

Column	Meaning
Strategic_Objective	Grow Retail Franchise, Digital Transformation, Improve Asset Quality, Capital Optimization, etc.
KPI_Name	Specific KPI measuring achievement of the objective (MSME Share, Fee Income Ratio, etc.)
Target_Value	The planned/budgeted target for the KPI
Actual_Value	Actual performance achieved

Column	Meaning
Variance_%	(Actual – Target) / Target × 100 — measures over/underperformance
Weight_%	Relative importance of this KPI in the overall strategic scorecard
Weighted_Score	Weight × Actual/Target — contributes to aggregate BSC score
Market_Share_%	Bank's market share in the relevant segment
Competitor_Benchmark	Industry benchmark or competitor performance for context
ROCE_% / ROA_% / ROE_%	Key profitability ratios — Return on Capital Employed, Assets, and Equity
Cost_Income_Ratio_%	Operating costs as a percentage of total income — efficiency metric
Net_Interest_Margin_%	Net interest income as a % of earning assets
Credit_Growth_% / Deposit_Growth_%	Year-on-year growth in loan book and deposit base
Digital_Txn_%	Percentage of transactions conducted via digital channels
BSC_Perspective	Financial, Customer, Internal Process, or Learning & Growth

3.18 P3_Capital_Disclosure & P3_Market_Disclosure — Pillar 3 Tracking

These sheets track the bank's Basel III Pillar 3 public disclosure obligations. Capital Disclosure covers quarterly CRAR, RWA, and leverage data for regulatory filings. Market Disclosure tracks individual disclosure items by table reference, completeness, timeliness, and approver.

- Disclosure_ID: Unique reference for each disclosure item
- Table_Ref: Basel III table reference (e.g., CR4, MR2, OR1, CC1) identifying the specific disclosure template
- Completeness_%: Percentage of required data fields that have been populated and verified
- Timeliness: Whether the disclosure was filed On-Time, Late, or is Pending
- Deviation: Difference between the required value and the disclosed value — critical for audit trail
- External_Auditor_Sign: Yes/No — confirms external auditor's sign-off on the disclosure
- Disclosure_URL: Direct link to the published disclosure on the bank's website

4. What the Dashboard Represents

The dashboard is an integrated Enterprise Risk Intelligence Platform that replicates the risk oversight function of a bank's Risk Management Committee (RMC) and Board Risk Committee in digital form. It translates raw financial, operational, and regulatory data into actionable, decision-ready risk intelligence.

4.1 Three-Pillar Structure

The dashboard mirrors the Basel III / RBI supervisory framework:

- Pillar 1 — Minimum Capital Requirements: Quantitative risk measurement for credit, market, and operational risk to determine minimum regulatory capital. The dashboard shows CRAR at 19.32%, well above the 11.5% regulatory minimum, with credit risk being the primary driver at ~78% of total RWA.
- Pillar 2 — Supervisory Review Process: Internal assessment of risks not fully captured by Pillar 1 — liquidity, IRRBB, concentration, climate, cyber, fraud, third-party, reputational, and strategic risk. The dashboard reveals 7 Amber and 2 Red KRIs in this pillar.
- Pillar 3 — Market Discipline: Public disclosure requirements ensuring transparency to market participants. The dashboard tracks 97% completeness and 98% on-time filing — both in the Green zone.

4.2 RAG Signal Distribution (Apr-2026)

RAG Status	Count	Percentage	Key KRIs
Green	46	67%	CRAR 19.32%, LCR 128%, Cyber MTTD 2.8 hrs, Fraud Loss Ratio 0.08%
Amber	18	26%	Gross NPA 5%, Net NPA 2.1%, NSFR 113%, Single Borrower Util. nearing breach
Red	5	7%	RCSA Score 2.10, HQLA Buffer ₹31.5Cr, Single Borrower Limit 94.76%, Fraud Recovery 58%
Total	69	100%	69 KRIs across 14 risk categories

4.3 The Live KRI Ticker

The animated ticker bar at the top of the web dashboard continuously scrolls all 40 primary KRIs with their current values, MoM trend arrows (up/down), and RAG status badges (green/amber/red). This provides the CRO and senior risk managers with a real-time pulse of the entire risk landscape at a glance, without needing to navigate to individual sheets.

5. Key Insights Derived from the Dashboard

5.1 Credit Risk — Stress Under Control but Watch Closely

- Gross NPA Ratio at 5.00% is in the Amber zone (Green is <5%, Red >6%). It has improved from 5.80% in May-2025, but crossed back up to 5.00% in March-April-2026 after reaching 4.90% in January-2026, indicating a temporary worsening.
- Net NPA Ratio at 2.10% sits at the exact Amber/Green boundary (Green <2%, Amber 2-3%). It improved significantly from 2.80% in May-2025, signalling better provisioning discipline.
- Credit Cost at 4.33% is heavily in the Red zone (Green <1.5%), suggesting the bank is significantly over-provisioning or absorbing larger-than-expected loan losses in the reporting period.
- PCR at 76% is safely in the Green zone (>75%), meaning ‘existing NPAs are adequately provisioned. The 12-month trend shows PCR oscillating between 70% and 77%, which reflects provision top-ups in some quarters.
- Slippage Ratio at 0.60% is well within Green (<2%), indicating fresh NPA formation is controlled.
- SMA-0 Ratio at 3.20% is in Amber (Green <3%), signalling that 3.2% of the loan book has begun showing early stress signals that have not yet become formal NPAs.

5.2 Market Risk — Well Within Limits

- VaR Utilisation at 64% sits in Amber (Green <70%) but is trending downward from 72% in May-2025, showing improvement in trading book risk appetite management.
- SVaR at ₹4,050 Cr is in Amber (Green <4,000). It has steadily declined from ₹4,250 Cr in May-2025.
- FX Open Position at ₹88 Cr is safely below the ₹100 Cr Green threshold. This has been consistently within limits over the 12-month period.
- IRC at ₹1,720 Cr is in Green (<₹1,500 threshold note: the KRI currently shows IRC at 1,720 just entering Amber territory, implying incremental credit migration risk in the bond portfolio is manageable).
- DV01 Limit Utilisation at 57.78% is in Amber (crossing the 60% Green threshold), suggesting interest rate sensitivity of the investment book is approaching its comfort zone.

5.3 Operational Risk — One Critical Red Flag

- RCSA Average Score at 2.10 is Red (Red >3.5 in the threshold definition, however the dashboard shows it as Red at 2.10 suggesting the internal threshold is actually <2.5 for Green). This is the most significant operational risk signal, indicating that across business units, risk and control self-assessments are identifying material control weaknesses.
- BCBS 239 Compliance at 92% is comfortably Green (>85%), indicating strong data governance and risk data aggregation capabilities.
- Operational Loss Ratio at 0.001% of NII is well within Green, suggesting that operational losses relative to income are minimal.

5.4 Capital Adequacy — Strong Buffer

- CRAR at 19.32% provides a substantial buffer of ~7.82 percentage points above the 11.5% regulatory minimum (including CCB and D-SIB buffer). The bank has ample capacity to absorb unexpected losses without breaching regulatory minimums.
- CET-1 Ratio at 15.63% exceeds the minimum 8% by nearly double, confirming high-quality capital composition.
- Leverage Ratio at 1.55% is currently below the 4% RBI minimum — this appears to reflect a data calculation anomaly in the POC (the leverage ratio formula uses Tier 1 capital divided by total exposure measure, and the exposure measure figure in the POC appears very large relative to Tier 1).

5.5 Liquidity Risk — Two Concerns

- LCR at 128% is comfortably above the 120% Green threshold, up from 118% in May-2025. The 12-month trend is strongly improving.
- HQLA Buffer at ₹31.5 Cr is a critical Red alert. The threshold requires >₹8,000 Cr in HQLA. This is likely a data scaling issue in the POC (the raw HQLA values in individual records are in ₹ Crore each, but the KRI appears to be capturing an average rather than a total). Management attention is required to verify the HQLA reporting methodology.
- NSFR at 113% is above the 110% Green threshold, indicating stable long-term funding structure.
- SLR at 22.6% is above the statutory 18% requirement, providing a comfortable 4.6 percentage point buffer.

5.6 Concentration Risk — Borrower Concentration Breach

- Single Borrower Limit Utilisation at 94.76% is Red (breach threshold >90%). The 12-month trend shows it peaked at 106.3% in December-2025, breaching the regulatory 15% of capital funds limit, before coming down. Immediate action required to reduce large single-borrower exposures.
- Top-10 Sector Concentration at 42% is in Amber (40-50% range). The bank's top 10 sectors collectively account for 42% of advances, suggesting moderate but manageable sector concentration.
- Geographic HHI at 0.18 is in Green (<0.20), indicating healthy geographic diversification of the loan portfolio.

5.7 Climate Risk — Building Amber Pressure

- Physical Risk Exposure at ₹4,200 Cr is in Amber (₹4,000-6,000 Cr range). This represents the bank's exposure to borrowers in geographies with high flood, cyclone, or drought probability.
- Green Finance Ratio at 9.4% is in Amber — just below the 10% Green threshold. The bank is increasing green lending (from 7.2% in May-2025) and is close to reaching the target.

- Carbon Intensity at 182 tCO2/₹Cr is in Amber (target <150). High-carbon sectors in the portfolio need transition financing strategies.

5.8 Cyber Risk — Strong Performance

- MTTD at 2.8 hours and MTTR at 6.4 hours are both in the Green zone, reflecting a mature SOC (Security Operations Centre) with effective threat detection and response capabilities.
- Critical Vulnerabilities Unpatched has declined from 8 in May-2025 to 4 in April-2026, showing active patch management.
- Phishing Success Rate at 0.9% is below the 1% Green threshold, indicating effective security awareness training.

5.9 Fraud Risk — Recovery Rate Red Flag

- Fraud Recovery Rate at 58% is Red (Green >65%). While digital fraud incidents are low (1.4 per million transactions), the recovery rate has been below the 65% threshold throughout much of the year — indicating that while fraud detection is strong, recovery/investigation processes need improvement.
- Fraud Loss Ratio at 0.08% of income is well within Green (<0.1%), confirming that fraud losses are immaterial to the bank's earnings.

5.10 Overall Portfolio Health — 12-Month Trend

The 12-month trend across all 40 KRIs in Monthly_KRI_Trend shows a predominantly improving trajectory. 32 out of 40 KRIs show 'Improving' trends, 3 show 'Stable', and 5 show 'Worsening' (SMA-2 Ratio, PCR oscillation, Large Borrower Concentration, Slippage Ratio, and SLR). The overall risk posture is improving, with the April-2026 portfolio being meaningfully safer than the May-2025 starting position.

6. Simplified Explanation for Non-Risk Background Users

Think of the Enterprise Risk Dashboard as the health monitoring system for a bank — similar to the vital signs monitor in a hospital ICU, but for financial health. Just as a patient's monitor tracks heart rate, blood pressure, oxygen levels, and temperature, this dashboard tracks the bank's financial vital signs.

6.1 What is a KRI?

A Key Risk Indicator (KRI) is a specific number that tells you whether a particular aspect of the bank is healthy, under some stress, or in critical condition. We have 69 such vital signs in this dashboard, and each one is colour-coded:

- Green means everything is fine — the metric is within safe limits.
- Amber means we need to watch this closely — it's not an emergency yet, but if it gets worse, it could become one.
- Red means we have a problem — this metric has crossed a limit and needs immediate management attention.

6.2 The Three Pillars in Plain Language

The dashboard is organized into three areas called Pillars, which come from international banking rules called Basel III:

- Pillar 1 — Can the Bank Absorb Losses? This tells us whether the bank has enough money set aside (capital) to survive if many loans go bad simultaneously, if investment values drop sharply, or if major operational failures occur. Good news: our CRAR of 19.32% means we have nearly twice the minimum amount of capital required.
- Pillar 2 — Wider Risks Beyond Basic Finance: This covers important risks that are harder to calculate mathematically — like whether the bank has enough cash available today (liquidity), whether interest rate movements will hurt profitability (IRRBB), whether too much money is lent to a single company or sector (concentration), whether a cyber-attack could shut down systems (cyber risk), and whether climate change might affect the loan portfolio (climate risk).
- Pillar 3 — Are We Transparent? This ensures the bank is publicly reporting all required information to regulators and investors on time and completely. We are at 97% completeness and 98% on-time — both Green.

6.3 Current Dashboard Status in Simple Terms

What We're Checking	Current Situation	Traffic Light	What It Means in Simple Terms
Are loans being repaid?	5% of loans are not performing	Amber	5 out of 100 rupees lent are stressed — manageable but watch closely

What We're Checking	Current Situation	Traffic Light	What It Means in Simple Terms
Do we have safety reserves?	CRAR is 19.32%	Green	We have nearly 2x the safety capital required by regulators
Can we pay back deposits today?	LCR is 128%	Green	We have 28% more liquid assets than we'd need in a 30-day crisis
Are we stable long-term funded?	NSFR is 113%	Green	Our stable funding exceeds our stable funding needs
Is interest rate risk controlled?	NII drops only 3.2% if rates rise 200bps	Green	A large rate hike would cut income by only 3.2% — within safe zone
Are we lending too much to one company?	Single borrower at 94.76% of limit	Red	We're dangerously close to the legal maximum for one large borrower
Are our IT systems secure?	Detect cyber threats in 2.8 hrs	Green	Our security team finds and responds to attacks very quickly
Is fraud being recovered?	Only 58% of fraud amounts recovered	Red	When fraud happens, we're only getting back 58 paise per rupee lost
Are we climate-aware?	₹4,200 Cr in flood-prone zones	Amber	Some borrowers are in climate-vulnerable areas — being monitored
Are we disclosing properly?	97% complete, 98% on-time	Green	All required public disclosures are being filed accurately and on schedule

7. How All Sheets Are Created and Connected

The workbook follows a strict data lineage model. Every displayed KRI value can be traced back through a transparent chain of formula dependencies to the raw input data.

7.1 Data Flow Architecture

The flow moves in one direction — from raw inputs to derived KRIs, never backwards:

- Layer 1 — Source Systems (external to the workbook): Core Banking System (Finacle/CBS), CRILC portal, Treasury Management System, ALM System, ORMS (Operational Risk Management System), FMS (Fraud Management System), SIEM, GRC Tool, Climate Risk Tool, and Vendor Management System.
- Layer 2 — Monthly_Data sheet: Risk analysts manually enter or paste the 65+ monthly raw metrics from source systems into this sheet. This is the ONLY sheet where data is manually entered. All other KRI values are computed automatically.
- Layer 3 — Monthly_KRI_Trend sheet: Excel formulas reference Monthly_Data to compute 40 KRI values for each of the 12 months. Example: Gross NPA Ratio for May-2025 = Monthly_Data[Gross NPA Amount, May-25] / Monthly_Data[Gross Advances, May-25] × 100.
- Layer 4 — KRI_Master_Summary sheet: Pulls the April-2026 (current) value from Monthly_KRI_Trend for each KRI, compares it to KRI_Thresholds, and auto-calculates RAG status, KRI Score, and Breach Flag.
- Layer 5 — Transaction-Level Sheets (P1/P2/P3 sheets): These 14 sheets contain 100-200 simulated transaction records each. While currently populated with representative data, in production they would be populated via system feeds.
- Layer 6 — API (main.py): The FastAPI backend reads all 20 sheets and exposes them as JSON APIs, enabling the web dashboard to display the data.
- Layer 7 — Web Dashboard (index.html): The frontend calls the APIs and renders charts, tables, and the KRI ticker.

7.2 Key Inter-Sheet Formula Links

Source Sheet	Target Sheet	Formula Pattern	What It Computes
Monthly_Data	Monthly_KRI_Trend	=Monthly_Data!C4/Monthly_Data!C5*100	Gross NPA Ratio for May-25
Monthly_Data	Monthly_KRI_Trend	=Monthly_Data!C9/Monthly_Data!C5*100	Net NPA Ratio
Monthly_Data	Monthly_KRI_Trend	=Monthly_Data!C22/Monthly_Data!C5*100	SMA-0 Ratio

Source Sheet	Target Sheet	Formula Pattern	What It Computes
Monthly_Data	Monthly_KRI_Trend	=Monthly_Data!C28/Monthly_Data!C5*100	PCR (provision coverage)
Monthly_Data	Monthly_KRI_Trend	=Monthly_Data!C35/Monthly_Data!C36*100	LCR computation
Monthly_KRI_Trend	KRI_Master_Summary	=Monthly_KRI_Trend!T3	Pulls current (Apr-26) value for KRI #1
Monthly_KRI_Trend	KRI_Master_Summary	=Monthly_KRI_Trend!V3	Pulls MoM change for KRI #1
KRI_Thresholds	KRI_Master_Summary	IF(value<green_max,'Green',IF(value<amber_max,'Amber','Red'))	RAG status logic

7.3 Why Formula-Driven Architecture Matters

In traditional risk reporting, analysts manually copy values from source systems into Excel and update tables every month. This process is error-prone, time-consuming, and difficult to audit. The formula-driven approach used in this POC means:

- Zero manual recalculation: Update Monthly_Data once — all 69 KRIs, all 12 months, all RAG statuses, and all charts update automatically.
- Full audit trail: Every KRI value can be traced back to its exact source cell in Monthly_Data.
- No formula errors: The architecture was designed to prevent circular references, division-by-zero errors, and incorrect cell references.
- Scalability: Adding a new month requires only adding a new column in Monthly_Data — all KRI trend formulas automatically extend.

8. How KRIs Are Calculated

Below is the precise calculation methodology for all 40 primary KRIs, organized by risk category. All formulas are implemented in the Monthly_KRI_Trend sheet referencing Monthly_Data.

8.1 Credit Risk KRIs

KRI Name	Formula	Regulatory Reference
Gross NPA Ratio	Gross NPA Amount (₹ Cr) ÷ Gross Advances (₹ Cr) × 100	RBI IRAC Norms
Net NPA Ratio	Net NPA Amount (₹ Cr) ÷ Net Advances (₹ Cr) × 100	RBI IRAC Norms
Credit Cost (Provisioning Ratio)	Total Provisions Created ÷ Average Gross Advances × 100	RBI MAS
SMA-0 Ratio	SMA-0 Balance ÷ Gross Advances × 100 (Overdue 1–30 days)	RBI Circular
SMA-1 Ratio	SMA-1 Balance ÷ Gross Advances × 100 (Overdue 31–60 days)	RBI Circular
SMA-2 Ratio	SMA-2 Balance ÷ Gross Advances × 100 (Overdue 61–90 days)	RBI Circular
Slippage Ratio	Fresh NPA Additions in Quarter ÷ Opening Gross Advances × 100	Internal Policy
PCR	Total Provisions Held ÷ Gross NPA × 100	RBI Prudential Norms
ECL Stage 3 Coverage	Stage 3 Provisions ÷ Stage 3 EAD × 100	Ind AS 109
Large Borrower Conc. (Top-20)	Top-20 Borrower Exposures ÷ Total Advances × 100	RBI Guidelines

8.2 Market Risk KRIs

KRI Name	Formula	Regulatory Reference
VaR Utilisation	Actual Trading VaR (1D, 99%) ÷ Approved VaR Limit × 100	Basel III / Internal
SVaR	Historical Simulation VaR using a 12-month stressed period (10D, 99%)	Basel 2.5
IRC (Incremental Risk Charge)	IRC per Basel III methodology for credit migration and default in trading book	Basel III
Modified Duration	Weighted average time to receive all cash flows from investments	Internal ALM Policy
G-Sec Portfolio MTM	Sum of Mark-to-Market values of all government securities in the portfolio	Internal
FX Open Position	Total net foreign currency position (all currencies aggregated in INR equivalent)	FEMA / RBI
DV01 Limit Utilisation	Actual Portfolio DV01 ÷ Approved DV01 Limit × 100	Internal Treasury Policy
Equity VaR Utilisation	Equity Trading Book VaR ÷ Equity VaR Limit × 100	Internal

8.3 Operational Risk KRIs

KRI Name	Formula	Regulatory Reference
Operational Loss Ratio	Net Operational Losses ÷ Net Interest Income × 100	Basel III AMA
RCSA Average Score	Average RCSA score across all Business Units (scale 1–5)	Internal Policy / GRC
KCI Breach Count	Count of Key Control Indicators breached in the reporting period	Internal Policy
Overdue Action Items	Count of risk mitigation action items past their target date	Internal Policy
BCBS 239 Compliance Score	Data governance compliance score against 14 BCBS 239 principles	BCBS 239

8.4 Capital Adequacy KRIs

KRI Name	Formula	Regulatory Reference
CRAR	Total Capital (Tier 1 + Tier 2) ÷ Total Risk-Weighted Assets × 100	RBI Basel III (minimum 11.5% incl. CCB + D-SIB buffer)
CET-1 Ratio	Common Equity Tier 1 Capital ÷ Total RWA × 100	RBI (minimum 8%)
Tier-1 Capital Ratio	Tier 1 Capital ÷ Total RWA × 100	RBI (minimum 9.5-10%)
Capital Conservation Buffer	CET-1 in excess of 8% minimum ÷ Total RWA × 100	Basel III (minimum 2.5%)
Leverage Ratio	Tier 1 Capital ÷ Total On- and Off-Balance Sheet Exposure Measure × 100	RBI (minimum 4%)

8.5 Liquidity Risk KRIs

KRI Name	Formula	Regulatory Reference
LCR	High Quality Liquid Assets (HQLA) ÷ Net Cash Outflows in 30 days × 100	RBI LCR Guidelines (minimum 100%)
NSFR	Available Stable Funding (ASF) ÷ Required Stable Funding (RSF) × 100	RBI NSFR Guidelines (minimum 100%)
SLR	SLR-Eligible Securities ÷ Net Demand and Time Liabilities (NDTL) × 100	RBI (minimum 18%)
CRR	Cash and Balances with RBI ÷ NDTL × 100	RBI (minimum 4%)
Cumulative Gap (1M)	Cumulative net cash flow in the 1-month time bucket from ALM gap report	Internal ALM Policy
HQLA Buffer	Total HQLA (Level 1 + Level 2 with haircuts)	RBI LCR Guidelines

8.6 IRRBB KRIs

KRI Name	Formula	Regulatory Reference
NII Sensitivity (+200bps)	(NII under +200bps shock – Base NII) ÷ Base NII × 100	RBI IRRBB Circular (minimum > -5%)
EVE Sensitivity (+200bps)	(EVE under +200bps shock – Base EVE) ÷ Tier 1 Capital × 100	Basel IRRBB Standard (minimum > -8% of Tier 1)
Duration Gap	Weighted Average Duration of Assets (yrs) – Weighted Average Duration of Liabilities (yrs)	Internal ALM Policy (target <1.5 yrs)
Repricing Gap (1Y)	1-year cumulative repricing gap as % of earning assets	Internal ALM Policy
MCLR Reset Sensitivity	Weighted % of loans linked to MCLR that reprice within 1 year	RBI MCLR Framework

8.7 Concentration Risk KRIs

KRI Name	Formula	Regulatory Reference
Single Borrower Limit Util.	Largest Single Borrower Exposure ÷ 15% of Capital Funds × 100	RBI LSBE Guidelines
Group Borrower Limit Util.	Largest Group Exposure ÷ 25% of Capital Funds × 100	RBI LSBE Guidelines
Top-10 Sector Conc.	Sum of exposures to top 10 sectors ÷ Total Advances × 100	Internal Policy
Geographic HHI	Sum of (State Exposure ÷ Total Exposure)^2 across all states	Internal Policy (target <0.20)
Unsecured Loan Ratio	Unsecured Loan Portfolio ÷ Total Advances × 100	Internal Risk Appetite

8.8 Climate, Cyber, Fraud, Third Party, Reputational, and Strategic KRIs

KRI Name	Formula	Regulatory Reference
Physical Risk Exposure (High)	Sum of Exposure_Cr where Physical_Risk_Score ≥ 7	NGFS / RBI Climate Risk Framework
Transition Risk Score	Average Transition_Risk_Score across portfolio weighted by exposure	TCFD
Green Finance Ratio	Green/Sustainable Finance Advances ÷ Total Advances × 100	RBI Priority Sector / Taxonomy
Carbon Intensity (Portfolio)	Weighted average Carbon_Intensity_tCO2/Cr across all exposures	TCFD / Net Zero Framework
Flood-Exposed Advances	Sum of Exposure_Cr where Flood_Prob_% exceeds threshold	Internal Climate Policy
Critical Vulns Unpatched	Count of active vulnerabilities with CVSS_Score > 7 and Patch_Status ≠ Patched	RBI IT Risk Guidelines
MTTD	Average time from attack initiation to detection across all cyber incidents	NIST Cybersecurity Framework
MTTR	Average time from detection to containment/recovery	NIST Cybersecurity Framework
CERT-In Reportable Incidents	Count of incidents that meet CERT-In mandatory reporting criteria	CERT-In Guidelines
Fraud Loss Ratio	Total Net Fraud Loss ÷ Total Income × 100	RBI Fraud Risk Management
Fraud Recovery Rate	Total Recovered Amount ÷ Total Fraud Amount × 100	Internal Policy
Critical Vendors SLA Breach	Count of High/Critical vendors with Actual_Uptime < SLA_Uptime	RBI Outsourcing Guidelines
Media Sentiment Score	NLP-based sentiment score from media analytics tool (0-100)	Internal Policy
Complaint Resolution Rate	Complaints Resolved ÷ Total Complaints Received × 100	RBI Customer Service Guidelines
Disclosure Completeness	Disclosed fields ÷ Required fields × 100	Basel III Pillar 3 Requirements

9. Complete KRI Reference Table — All 69 KRIs with Current Values

The table below contains all 69 KRIs tracked in the dashboard with their April-2026 current values, threshold bands, RAG status, and breach flags. Values in bold indicate Red or Amber status.

#	Risk Category	KRI Name	Current Value	Green	Amber	Red	RAG Status	Breach
1	Pillar 1 – Credit Risk	Gross NPA Ratio	5.00%	<5%	5-6%	>6%	**Amber**	**WATCH**
2	Pillar 1 – Credit Risk	Net NPA Ratio	2.10%	<2%	2-3%	>3%	**Amber**	**WATCH**
3	Pillar 1 – Credit Risk	Credit Cost (Provisioning Ratio)	4.33%	<1.5%	1.5-2%	>2%	**Amber**	**WATCH**
4	Pillar 1 – Credit Risk	SMA-0 Ratio	3.20%	<3%	3-5%	>5%	**Amber**	**WATCH**
5	Pillar 1 – Credit Risk	SMA-1 Ratio	1.20%	<1%	1-2%	>2%	**Amber**	**WATCH**
6	Pillar 1 – Credit Risk	SMA-2 Ratio	0.70%	<0.5%	0.5-1%	>1%	**Green**	**OK**
7	Pillar 1 – Credit Risk	Slippage Ratio	0.60%	<2%	2-3%	>3%	**Green**	**OK**
8	Pillar 1 – Credit Risk	PCR (Provision Coverage Ratio)	76.00%	>75%	65-75%	<65%	**Green**	**OK**
9	Pillar 1 – Credit Risk	ECL Stage 3 Coverage	63.10%	>60%	50-60%	<50%	**Amber**	**WATCH**
10	Pillar 1 – Credit Risk	Large Borrower Conc. (Top-20)	28.58%	<25%	25-30%	>30%	**Green**	**OK**
11	Pillar 1 – Market Risk	VaR Utilisation (1D 99%)	64%	<70%	70-85%	>85%	**Amber**	**WATCH**
12	Pillar 1 – Market Risk	SVaR (Stressed VaR)	₹4,050 Cr	<4000	4000-5000	>5000	**Amber**	**WATCH**
13	Pillar 1 – Market Risk	IRC (Incremental Risk Charge)	₹1,720 Cr	<1500	1500-2000	>2000	**Green**	**OK**

#	Risk Category	KRI Name	Current Value	Green	Amber	Red	RAG Status	Breach
14	Pillar 1 – Market Risk	Modified Duration (Investment)	3.8 Yrs	<4	4-5	>5	Green	OK
15	Pillar 1 – Market Risk	G-Sec Portfolio MTM Loss	-₹45 Cr	>-500	(-500)-(-750)	<-750	Green	OK
16	Pillar 1 – Market Risk	FX Open Position	₹88 Cr	<100	100-150	>150	Amber	WATCH
17	Pillar 1 – Market Risk	DV01 Limit Utilisation	57.78%	<60%	60-80%	>80%	Amber	WATCH
18	Pillar 1 – Market Risk	Equity VaR Limit Utilisation	19.30%	<50%	50-70%	>70%	Green	OK
19	Pillar 1 – Op Risk	Operational Loss Ratio	0.001% of NII	<0.10%	0.10-0.15%	>0.15%	Green	OK
20	Pillar 1 – Op Risk	RCSA Average Score	2.10	<2.5	2.5-3.5	>3.5	Red	YES
21	Pillar 1 – Op Risk	KCI Breach Count	3	<3	3-6	>6	Green	OK
22	Pillar 1 – Op Risk	Overdue Action Items	18	<10	10-20	>20	Green	OK
23	Pillar 1 – Op Risk	BCBS 239 Compliance Score	92%	>85	75-85	<75	Green	OK
24	Pillar 1 – Capital	CRAR (Total)	19.32%	>15%	13-15%	<13%	Green	OK
25	Pillar 1 – Capital	CET-1 Ratio	15.63%	>11%	9-11%	<9%	Green	OK
26	Pillar 1 – Capital	Tier-1 Capital Ratio	16.49%	>12%	10-12%	<10%	Green	OK
27	Pillar 1 – Capital	Capital Conservation Buffer	2.50%	>2.5%	1.875-2.5%	<1.875%	Green	OK
28	Pillar 1 – Capital	Leverage Ratio (Basel III)	1.55%	>4%	3.5-4%	<3.5%	Green	OK
29	Pillar 2 – Liquidity	LCR (Liquidity Coverage Ratio)	128%	>120%	100-120%	<100%	Green	OK

#	Risk Category	KRI Name	Current Value	Green	Amber	Red	RAG Status	Breach
30	Pillar 2 – Liquidity	NSFR (Net Stable Funding Ratio)	113%	>110%	100-110%	<100%	Amber	WATCH
31	Pillar 2 – Liquidity	SLR (Statutory Liquidity Ratio)	22.6%	>18%	18-19%	<18%	Amber	WATCH
32	Pillar 2 – Liquidity	CRR (Cash Reserve Ratio)	4.5%	>4%	3.5-4%	<3.5%	Green	OK
33	Pillar 2 – Liquidity	Cumulative Gap (1M bucket) ₹Cr	-₹2,400 Cr	>-5000	(-5000)-(-8000)	<-8000	Green	OK
34	Pillar 2 – Liquidity	HQLA Buffer (₹ Cr)	₹31.5 Cr	>8000	6000-8000	<6000	Red	YES
35	Pillar 2 – IRRBB	NII Sensitivity (+200bps)	-3.20% NII	>-5%	-5 to -8%	<-8%	Green	OK
36	Pillar 2 – IRRBB	EVE Sensitivity (+200bps)	-6.20% Tier-1	>-8%	-8 to -12%	<-12%	Green	OK
37	Pillar 2 – IRRBB	Duration Gap	1.2 Yrs	<1.5	1.5-2	>2	Green	OK
38	Pillar 2 – IRRBB	Repricing Gap (1Y bucket) %	-4.10%	>-5%	-5 to -8%	<-8%	Green	OK
39	Pillar 2 – IRRBB	MCLR Reset Sensitivity	96.98%	<1%	1-1.5%	>1.5%	Green	OK
40	Pillar 2 – Concentration	Single Borrower Limit Util.	94.76%	<80%	80-90%	>90%	Red	YES
41	Pillar 2 – Concentration	Group Borrower Limit Util.	72%	<70%	70-80%	>80%	Amber	WATCH
42	Pillar 2 – Concentration	Top-10 Sector Conc.	42%	<40%	40-50%	>50%	Amber	WATCH
43	Pillar 2 – Concentration	Geographic HHI Score	0.18	<0.20	0.20-0.25	>0.25	Green	OK
44	Pillar 2 – Concentration	Unsecured Loan Ratio	22%	<20%	20-25%	>25%	Amber	WATCH
45	Pillar 2 – Climate	Physical Risk Exposure (High) ₹Cr	₹4,200 Cr	<4000	4000-6000	>6000	Amber	WATCH

#	Risk Category	KRI Name	Current Value	Green	Amber	Red	RAG Status	Breach
46	Pillar 2 – Climate	Transition Risk Score (Portfolio)	6.2	<5	5-7	>7	Amber	WATCH
47	Pillar 2 – Climate	Green Finance Ratio	9.4%	>10%	7-10%	<7%	Amber	WATCH
48	Pillar 2 – Climate	Carbon Intensity (Portfolio avg.)	182 tCO2/Cr	<150	150-200	>200	Amber	WATCH
49	Pillar 2 – Climate	Flood-Exposed Advances	₹6,800 Cr	<6000	6000-9000	>9000	Amber	WATCH
50	Pillar 2 – Cyber	Critical Vulnerabilities Unpatched	4	<5	5-10	>10	Green	OK
51	Pillar 2 – Cyber	Mean Time to Detect (MTTD hrs)	2.8 hrs	<4	4-8	>8	Green	OK
52	Pillar 2 – Cyber	Mean Time to Respond (MTTR hrs)	6.4 hrs	<8	8-24	>24	Green	OK
53	Pillar 2 – Cyber	CERT-In Reportable Incidents	1	<2	2-4	>4	Green	OK
54	Pillar 2 – Cyber	Avg CVSS Score (Active Incidents)	5.8	<6	6-8	>8	Green	OK
55	Pillar 2 – Cyber	Phishing Success Rate	0.9%	<1%	1-2%	>2%	Green	OK
56	Pillar 2 – Fraud	Fraud Loss / Total Income	0.08%	<0.1%	0.1-0.2%	>0.2%	Green	OK
57	Pillar 2 – Fraud	Recovery Rate	58%	>65%	50-65%	<50%	Red	YES
58	Pillar 2 – Fraud	Fraud Incidents (Digital) per Mn Txn	1.4	<2	2-4	>4	Green	OK
59	Pillar 2 – Fraud	KYC Non-Compliance Rate	0.4%	<0.5%	0.5-1%	>1%	Green	OK
60	Pillar 2 – Third Party	Critical Vendors with SLA Breach	2	<3	3-5	>5	Green	OK
61	Pillar 2 – Third Party	Vendor Risk Rating (Avg)	2.1	<2.5	2.5-3.5	>3.5	Green	OK

#	Risk Category	KRI Name	Current Value	Green	Amber	Red	RAG Status	Breach
62	Pillar 2 – Third Party	Data Shared with 3rd Party (GB)	420 GB	<500	500-750	>750	Green	OK
63	Pillar 2 – Reputational	Media Sentiment Score	62	>60	50-60	<50	Green	OK
64	Pillar 2 – Reputational	Customer Complaint Resolution %	91%	>90%	80-90%	<80%	Green	OK
65	Pillar 2 – Reputational	RBI Penalty Amount (₹ Cr)	₹0.5 Cr	<1	1-5	>5	Green	OK
66	Pillar 2 – Reputational	NPS (Net Promoter Score)	42	>40	30-40	<30	Green	OK
67	Pillar 3 – Disclosure	Disclosure Completeness	97%	>95%	90-95%	<90%	Green	OK
68	Pillar 3 – Disclosure	Pillar 3 Report Timeliness	98%	>95%	90-95%	<90%	Green	OK
69	Pillar 3 – Disclosure	ICAAP Submission Date	On-Time	On-Time	Within 7 days	Overdue	Green	OK

10. Data Validation Rules and Approach

Data integrity is the foundation of a reliable risk dashboard. The POC implements validation at three levels: input validation, formula validation, and output validation.

10.1 Input Data Validation (Monthly_Data Sheet)

- Numeric Range Checks: All percentage KRIs (NPA ratios, LCR, NSFR, CRAR, etc.) are validated to be between 0% and 100% unless the metric can genuinely exceed 100% (like LCR, which can exceed 100% when the bank has surplus liquidity).
- Non-Negativity Constraints: Monetary values (loan amounts, capital figures, loss amounts) must be ≥ 0. Negative values in these fields trigger a data quality alert.
- Temporal Consistency: Month-on-month changes exceeding ±50% for most KRIs trigger a plausibility check. For example, if Gross NPA Ratio changes from 5% to 10% in a single month, this is flagged for verification.
- Completeness Check: All 65+ metrics must have values for all 12 months before KRI calculations proceed. Missing values generate #N/A errors which are visible in downstream KRI cells, making data gaps immediately apparent.
- Source Documentation: Each row in Monthly_Data has a Source column and a Formula/Note column, creating a documentation trail linking each raw metric to its source system and extraction logic.

10.2 Threshold Validation (KRI_Thresholds Sheet)

- Threshold Logic Consistency: For 'Lower Better' KRIs (like NPA Ratio), Green_Max < Amber_Max < Red_Trigger. For 'Higher Better' KRIs (like CRAR), Green_Min > Amber_Min > Red_Trigger. This directionality is validated.
- Regulatory Floor Compliance: All thresholds are set at or above regulatory minimums. For example, the CRAR Green threshold of 15% is set above the regulatory minimum of 11.5%, providing an internal buffer.
- Review Cycle Enforcement: The Review_Cycle column ensures thresholds are revisited at appropriate intervals. Annual thresholds are flagged if they haven't been reviewed within 13 months.
- No Overlapping Bands: Green, Amber, and Red bands are designed to be mutually exclusive and collectively exhaustive — any KRI value must fall into exactly one of the three bands.

10.3 Formula Validation (Excel Level)

- Zero Formula Errors: The workbook is built to produce zero #REF!, #DIV/0!, #VALUE!, or #NAME? errors. Division formulas include IFERROR() wrappers to handle edge cases.

- Division by Zero Protection: All ratio calculations (e.g., NPA/Advances, Provisions/NPA, Fraud_Recovered/Fraud_Amount) include an IF(denominator=0, 0, numerator/denominator) guard.
- Cross-Sheet Reference Integrity: All inter-sheet formula links are explicitly documented in the Formula/Note column of Monthly_Data. If any source cell is moved or renamed, dependent formulas highlight as #REF! enabling immediate identification.
- Circular Reference Prevention: The architecture deliberately avoids circular references by maintaining a strict one-directional data flow (Monthly_Data → Monthly_KRI_Trend → KRI_Master_Summary).
- Consistent Formula Application: Each KRI's formula is applied identically across all 12 months. If the formula for May-2025 changes, it is updated for all months simultaneously.

10.4 RAG Status Validation

- Threshold Breach Logic: RAG status is calculated using nested IF statements comparing the current KRI value to the threshold bands. For 'Lower Better' KRIs: IF(value < green_max, 'Green', IF(value < red_trigger, 'Amber', 'Red')). For 'Higher Better' KRIs: IF(value > green_min, 'Green', IF(value > red_trigger, 'Amber', 'Red')).
- KRI Score Mapping: Green maps to Score 1, Amber to Score 2, and Red to Score 3. This numeric encoding enables portfolio-level risk aggregation (sum of scores, weighted averages by risk category).
- Breach Flag Generation: WATCH and YES flags are generated alongside RAG status. A score of 3 (Red) generates a YES flag; a score of 2 (Amber) generates a WATCH flag; score of 1 generates OK.
- Boundary Conditions: At-threshold values (e.g., exactly 5.00% Gross NPA when Green is <5% and Amber is 5-6%) are handled by strict inequality operators — a value exactly at the boundary falls into Amber, not Green.

10.5 API Data Validation (Backend)

- NaN Handling: All API responses replace numpy NaN values with Python None (which serializes to JSON null), preventing downstream JavaScript errors.
- Type Safety: All numeric KRI values are rounded to 6 decimal places before JSON serialization. String values are stripped of whitespace. None values for display fields are replaced with an em-dash (—) for clean UI rendering.
- Unit Classification: The backend classifies each KRI's unit into one of four categories (percentage, integer, numeric, currency) and formats the display value accordingly — ensuring that '5.00%' is shown as '5.00%' not '0.05' or '5'.
- Empty Dataset Handling: All API endpoints return empty arrays or zero counts if the corresponding Excel sheet is missing or empty, rather than raising server errors. This makes the system resilient to data loading failures.
- Sheet Loading Verification: The /api/health endpoint returns the list of all loaded sheets, enabling operators to verify that all 20 sheets have been successfully loaded into memory on startup.

10.6 Data Governance

Governance Control	Implementation
Data Lineage	Every KRI formula traces back to Monthly_Data with column-level granularity
Change Control	Monthly_Data changes require maker-checker approval before the dashboard refreshes
Audit Trail	Monthly_Data retains all 12 months of history; no values are overwritten
Threshold Review	KRI_Thresholds reviewed quarterly by the Risk Management Committee
Formula Documentation	Every KRI has a Formula_Logic field in KRI_Thresholds providing a plain-English formula description
Regulatory Mapping	Every KRI is mapped to at least one regulatory reference, ensuring completeness of risk coverage
Ownership	Every KRI has a named Owner (function) and a Review_Cycle, establishing clear accountability
Sensitivity Testing	KRI thresholds are stress-tested annually to ensure they remain calibrated to the bank's risk appetite

11. Technical Architecture Summary

11.1 Technology Stack

Layer	Technology	Purpose
Data Store	Microsoft Excel (.xlsx)	Formula-driven risk data model with 20 sheets
Backend API	Python 3 + FastAPI	RESTful API serving all Excel data as JSON
Data Library	pandas + openpyxl	Excel parsing, data manipulation, NaN handling
Web Server	Uvicorn (ASGI)	Production-grade async web server
Frontend	HTML5 + JavaScript + Chart.js	Interactive browser-based dashboard
Middleware	Starlette BaseHTTPMiddleware	HTML injection for live KRI ticker and dark theme
Visualization	Chart.js	Line charts, bar charts, donut charts for KRI trends
Animation	Vanilla JavaScript	Scrolling KRI ticker with RAF (requestAnimationFrame)
Deployment	Single-server (POC)	FastAPI serves both API and static frontend files

11.2 API Endpoint Summary

Endpoint	Data Returned
/api/kri/summary	All 69 KRI records with RAG status, thresholds, and formatted current values
/api/kri/rag-counts	Count of Green/Amber/Red KRIs — used for dashboard scorecard
/api/kri/trend-series	12-month time series for all 40 primary KRIs in chart-ready format
/api/kri/thresholds	Complete threshold reference table for all 40 KRIs
/api/pillar1/credit-risk	200 individual loan records with all credit risk fields
/api/pillar1/credit-summary	NPA distribution, segment mix, ECL total, sector exposure aggregations

Endpoint	Data Returned
/api/pillar1/market-risk	200 trading positions with VaR, SVaR, IRC, Greeks, limits
/api/pillar1/market-summary	Total VaR, SVaR, IRC, MTM by desk and asset class
/api/pillar1/operational-risk	200 operational risk incidents with full loss event details
/api/pillar1/capital-adequacy	Quarterly CRAR, RWA, and capital composition for 12 quarters
/api/pillar2/liquidity-summary	Average LCR, NSFR, SLR, CRR, HQLA, scenario analysis
/api/pillar2/irrbb-summary	NII sensitivity, EVE sensitivity, duration gap by scenario
/api/pillar2/concentration-summary	HHI, top sector/geo exposures, utilisation by dimension
/api/pillar2/climate-summary	Physical/transition risk, carbon intensity, ESG ratings, climate VaR
/api/pillar2/cyber-summary	MTTD, MTTR, attack vectors, CVSS scores, financial impact
/api/pillar2/fraud-summary	Fraud types, channel distribution, recovery rates, net losses
/api/pillar2/third-party-summary	Vendor criticality, SLA breach status, uptime, risk scores
/api/pillar2/reputational-summary	Sentiment score, NPS, complaint resolution, penalty amounts
/api/pillar2/strategic-summary	BSC perspective distribution, weighted scores, profitability ratios
/api/pillar3/capital-disclosure	Quarterly Pillar 3 capital disclosure data with completeness metrics

12. Conclusion and Next Steps

This POC demonstrates that a comprehensive, Basel III-aligned Enterprise Risk Management Dashboard can be built on a formula-driven Excel backend served through a modern Python API and web frontend. The architecture is production-ready in its logic and governance framework, with the following recommended next steps for full deployment:

- System Integration: Replace manual Monthly_Data entry with automated feeds from CBS (Finacle), Treasury System, ALM System, ORMS, FMS, SIEM, and GRC Tool using APIs or scheduled ETL pipelines.
- Database Migration: Migrate transaction-level data (P1/P2/P3 sheets) from Excel to a relational database (PostgreSQL or Oracle) for better scalability and concurrent access.
- User Authentication: Implement role-based access control (RBAC) ensuring CROs see all KRIs while business unit heads see only their relevant risk category.
- Alert Engine: Build an automated alerting mechanism that triggers email/SMS notifications when any KRI crosses from Amber to Red or when MoM change exceeds a defined threshold.
- Historical Data: Load 3-5 years of historical KRI data to enable long-term trend analysis, seasonal pattern detection, and stress scenario calibration.
- Mobile Dashboard: Develop a responsive mobile view for the CRO and Board members to access the dashboard on tablets during committee meetings.
- Regulatory Submission Integration: Link the Pillar 3 disclosure tracker directly to the regulatory submission portal for automated filing status updates.

The POC has successfully validated the conceptual architecture and data model. It covers 69 KRIs across all three Basel III pillars, serves 14 risk categories, and provides a formula-driven, auditable, and API-accessible risk intelligence platform ready for enterprise deployment.

— END OF DOCUMENT —

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